Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering 09/2023 – Present

Kabataş Boys' High School 09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.

Experience

Executive Board Member & DJ at Radyo Boğaziçi 09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant 06/2025 – 09/2025

- Gained international work experience in the USA, improving English communication, teamwork, and adaptability in a multicultural environment.

Regional Organizational Team Member at ESTIEM 09/2023 – Present

- Contributed to the planning, coordination of events, and organizational processes within ESTIEM, an international network for Industrial Engineering and Management students.

Technical Projects (@GitHub)

FinTech & Trading Systems — Python, FastAPI, Next.js, PostgreSQL, Docker, WebSockets

- Built **AlphaTrigger** and an **AI-powered FinTech platform**, combining real-time trading simulation, volatility-adjusted position sizing, portfolio risk analysis, OpenAI-generated insights, and a live limit-order matching engine.

Enterprise Admin Dashboard — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

- Developed **Performo**, a real-time administrative dashboard with operational metrics, server-side synchronized state management, force-dynamic rendering, and optimized SQL indexing for stable monitoring workflows.

Systems & CPU Scheduling — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- Implemented **SchedSim**, a zero-standard-library CPU scheduling simulator supporting FCFS, SJF, SRTF, Priority, and Round Robin algorithms through direct Linux syscalls.

Interpreters & Language Design — C, Python, Lexer Design, Functional Paradigms

- Built **Spireslayer** and **Aura**, including a C-based backtracking command interpreter and a Python functional-language interpreter with closures, higher-order functions, and lexical/dynamic scoping.

Algorithms & Matching Engines — Java, Graph Theory, Heaps, Complexity Analysis

- Developed **MatrixNet** and **GigMatch-Pro**, implementing Tarjan's SCC, articulation point detection, open addressing optimization, and dynamic max-heap based task-matching algorithms.

Game & WebAssembly Engineering — Java, C++, Qt6, WebAssembly

- Built **Nightpass** and **Qt 2048**, including a Java rule-engine card game and a C++/Qt6 2048 clone cross-compiled to WebAssembly with optimized grid logic and undo state history.

Skills

- **Programming:** Python, Java, C/C++, Qt6, WebAssembly, TypeScript, JavaScript, SQL, x86-64 Assembly, Bash
- **Low-Level & Systems:** Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management, Qt Framework
- **Enterprise Full-Stack & AI:** Next.js 15+, React, Prisma, SQLAlchemy, FastAPI, Tailwind CSS, PostgreSQL, Advanced Agentic Coding
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Alembic, ccache, Pytest, CI/CD, Makefile, LaTeX, MS Office
- **Languages:** Turkish (native), English (fluent), German (elementary)

Volunteer & Extracurricular Experience

- **TOÇEV**: Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds. (09/2019 – 03/2020)
- **Boğaziçi University Industrial Engineering Club**: Actively contributed to the Volunteering Division. (09/2023 – 06/2024)
- **Kabataş Boys' High School CIP Club**: Organized and executed various community-oriented projects. (09/2018 – 06/2023)
- **Athletics**: Student-Athlete, Boğaziçi University Sailing Team and Ski Team & Windsurf Athlete, Competing Individually